

USD15/tCO2e CAN REPLICATE THE ~3% THESIS — BUT POLICY DESIGN DETERMINES WHETHER THE GDP TRADE-OFF PERSISTS

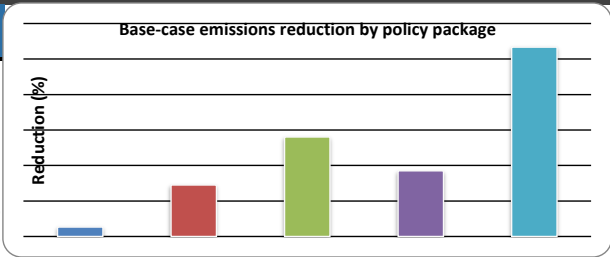
Executive readout: thesis replication, output ranges, and recommended package. Values update when blue inputs change.

KEY FINDINGS

- 1. The \$15 tax-only case produces 2.9% net reduction (26.9 MtCO2e), closely testing the article's 2.9% anchor.
- 2. Adding 5% efficiency and 30% recycling changes the modeled GDP impact from -0.8% to -0.1% while increasing emissions reduction to 5.6%.
- 3. Across five Base policy cases, net emissions reduction ranges from 0.5% to 10.7%.
- 4. A 30% recycling share directs \$3.7bn to transition support and reaches up to 50% of households under the article's compensation anchor.
- 5. Even the \$30 accelerated case delivers only 68% of the unconditional NDC reduction in this directional model; carbon pricing is a package component, not a standalone solution.

POLICY CASE COMPARISON — BASE SCENARIO

| Metric                         | Pilot ETS | \$15 Tax Only | \$15 Just Transition | \$15 Protected | \$30 Accelerated |
|--------------------------------|-----------|---------------|----------------------|----------------|------------------|
| Net emissions reduction (%)    | 0.5%      | 2.9%          | 5.6%                 | 3.7%           | 10.7%            |
| Reduction (MtCO2e)             | 4.9       | 26.9          | 52.0                 | 34.3           | 99.0             |
| Net GDP impact (%)             | 0.1%      | -0.8%         | -0.1%                | 0.2%           | -0.1%            |
| Net GDP impact (USD bn)        | 0.4       | (3.8)         | (0.5)                | 0.8            | (0.4)            |
| Net fiscal proceeds (USD bn)   | 0.2       | 12.8          | 12.4                 | 6.7            | 23.5             |
| Households compensated (%)     | 33%       | 0%            | 50%                  | 50%            | 50%              |
| NDC unconditional progress (%) | 3%        | 18%           | 36%                  | 23%            | 68%              |



RECOMMENDATION

Proceed with a staged \$15/tCO2e broad price only as a package: preserve the marginal price signal, recycle at least 30% of proceeds to lower-income households, and pair it with measurable energy-efficiency delivery. Treat the tax-only case as the counterfactual—not the recommended design.

Confidence Directional / Medium-Low Critical uncertainty Behavioural response and the macro benefit of recycling are not identified in the article.